Federated Optimization on MIMIC-IV ICU Mortality

Federated Learning Optimization Project

April 27, 2026

Abstract

This report extends the UCI HAR federated optimization project to MIMIC-IV ICU mortality prediction. The experiment keeps the same optimization core: FedAvg convergence, CVaR-style fairness, linear-programming communication shadow prices, and genetic hyperparameter search. The clinical task is more difficult than UCI HAR because it is binary, imbalanced, and naturally non-IID across ICU care units. The final cohort contains 73141 ICU stays, 1021 engineered features, and 9 federated ICU clients.

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1 Problem Definition

The goal is to predict in-hospital mortality for an ICU stay using information available in the first 24 hours after ICU admission. Each federated client is an ICU care unit identified by `first_careunit`. This is a natural non-IID split because MICU, SICU, CVICU, Neuro ICU, and other units treat different populations with different mortality rates.

2 Preprocessing and Cohort

The preprocessing stage joined `icustays`, `admissions`, and `patients` to define the cohort and label. It then joined chart events, lab events, input events, output events, procedure events, prescriptions, diagnoses, procedures, services, and transfers. Time-stamped clinical event tables were restricted to the first 24 hours after ICU admission to reduce leakage. Numeric events were aggregated into per-stay mean, min, max, standard deviation, and count features where appropriate.

Table 1: MIMIC-IV model matrix summary

rows	features	clients	positive_label	negative_label	mortality_rate	train_rows	test_rows	pr
73141.0000	1021.0000	9.0000	8340.0000	64801.0000	0.1140	54853.0000	18288.0000	

Table 2: Federated ICU client summary

client_id	client_name	rows	mortality_rate	deaths	
2.0000	Medical Intensive Care Unit (MICU)	15940.0000	0.1478	2356.0000	1358
3.0000	Medical/Surgical Intensive Care Unit (MICU/SICU)	12661.0000	0.1456	1844.0000	1081
0.0000	Cardiac Vascular Intensive Care Unit (CVICU)	11572.0000	0.0425	492.0000	1108
7.0000	Surgical Intensive Care Unit (SICU)	11192.0000	0.1188	1330.0000	986
8.0000	Trauma SICU (TSICU)	8668.0000	0.1039	901.0000	776
1.0000	Coronary Care Unit (CCU)	8319.0000	0.1285	1069.0000	725
4.0000	Neuro Intermediate	2015.0000	0.0199	40.0000	197
6.0000	Neuro Surgical Intensive Care Unit (Neuro SICU)	1757.0000	0.1633	287.0000	147
5.0000	Neuro Stepdown	1017.0000	0.0206	21.0000	99

3 EDA

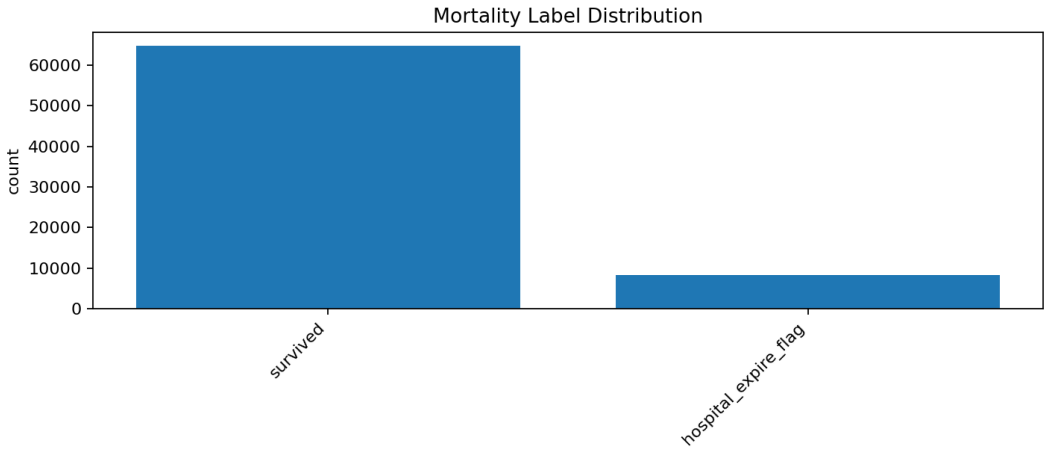

Figure 1: Mortality label imbalance in the final cohort.

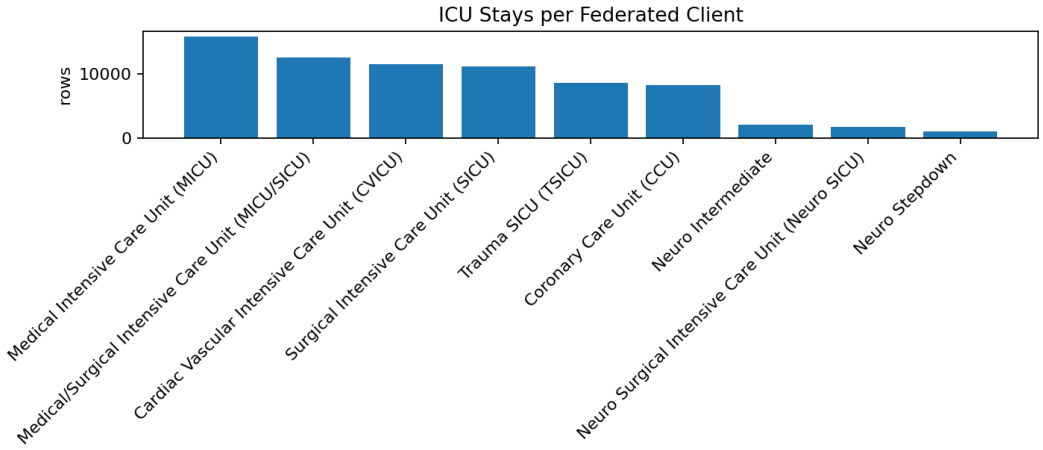

Figure 2: ICU stays per federated client.

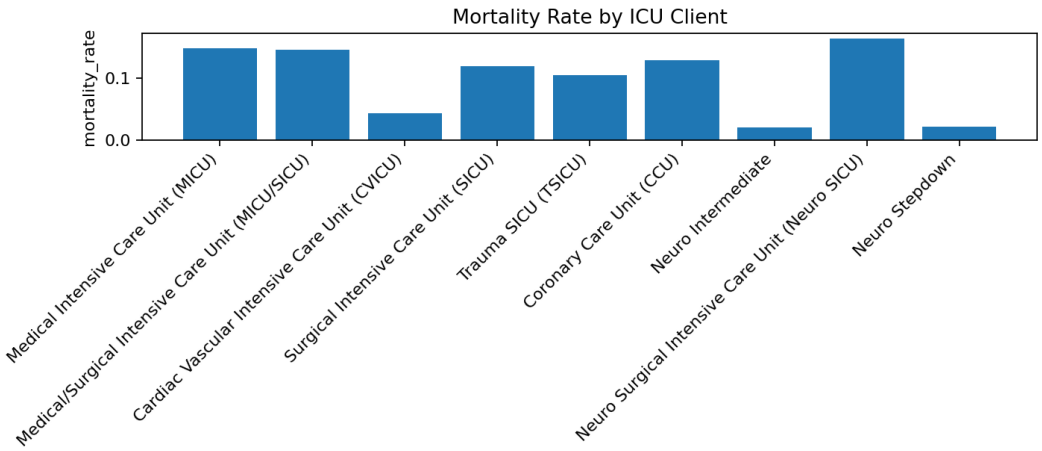

Figure 3: Mortality rate differs strongly by ICU care unit.

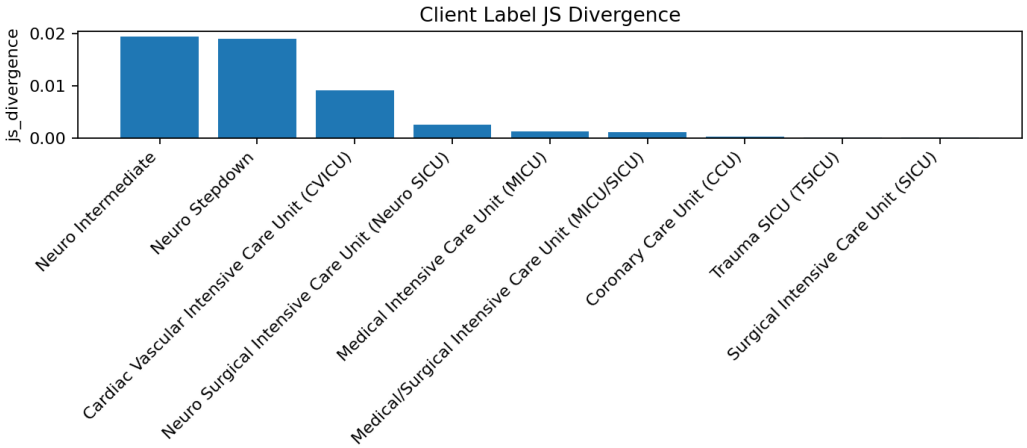

Figure 4: Non-IID label severity measured by Jensen-Shannon divergence.

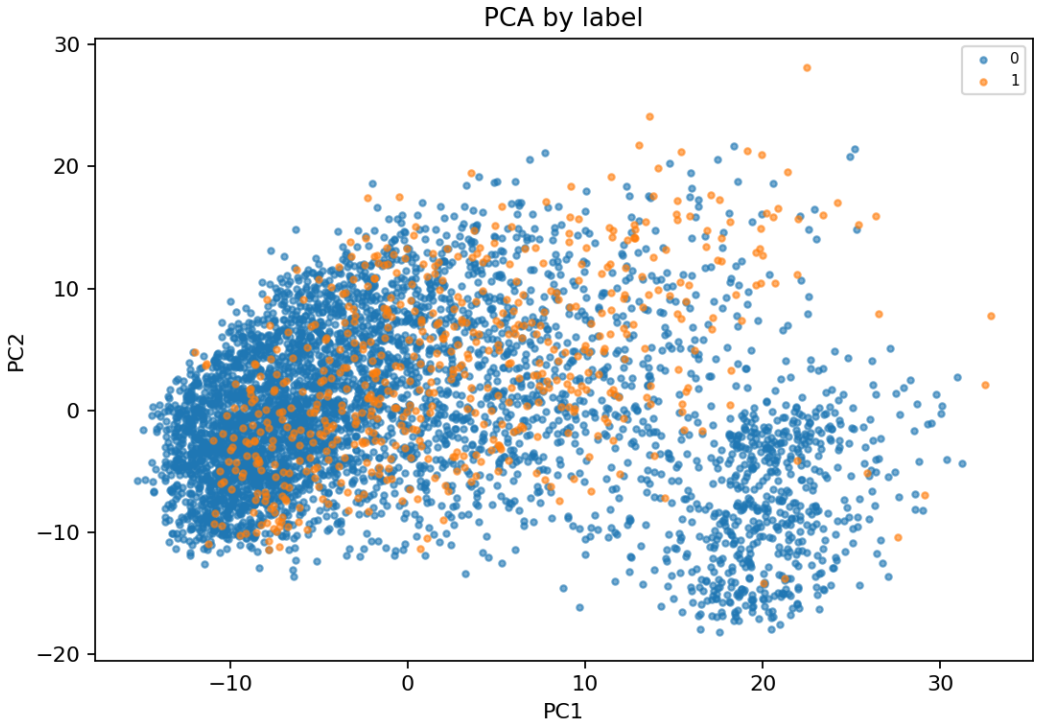

Figure 5: Two-dimensional PCA visualization by mortality label. PCA is used for EDA only, not as the main training representation.

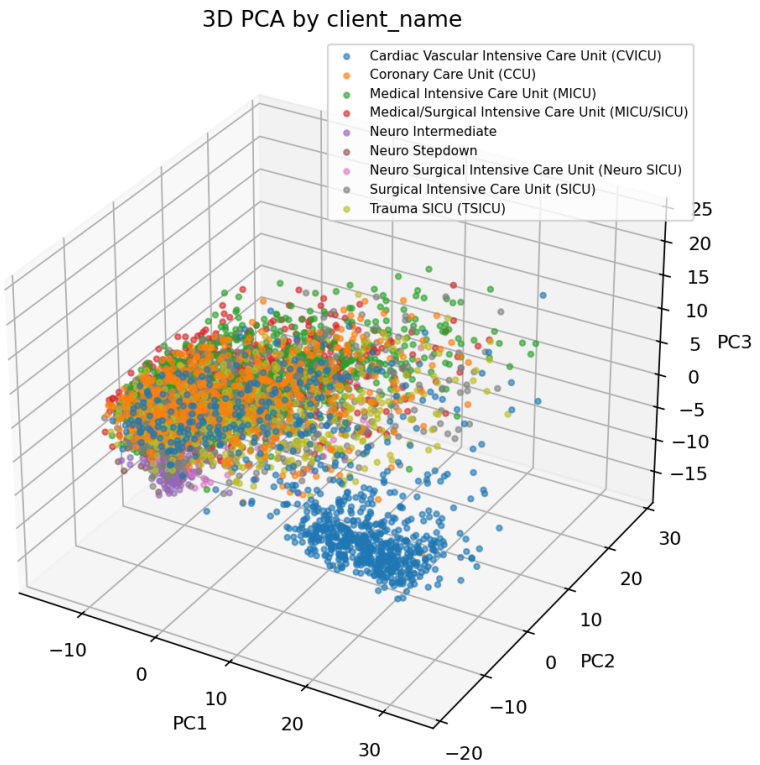

Figure 6: Three-dimensional PCA visualization by ICU client.

4 Training Configuration

All main validation uses ten seeds: `7,11,19,23,29,31,37,41,43,47`. The MIMIC model is a tabular MLP with 796 inputs and two output classes. Weighted cross entropy is used because mortality prevalence is about 11.4 percent. Early stopping uses loss with `max_rounds=1000`, `patience=25`, and `min_delta=0.0005`.

5 Main Results

The strongest mean AUPRC in this run is `fedavg_default` with mean AUPRC 0.6316. Accuracy is reported, but clinical interpretation focuses more on AUPRC, sensitivity, specificity, balanced accuracy, and worst-client metrics.

Table 3: Method comparison across seeds

method	n	final_accuracy_mean	final_auroc_mean	final_auprc_mean	final_balanced_accu
fedavg_default	10.0000	0.8776	0.8889	0.6316	
cvar_0	10.0000	0.8640	0.8860	0.6260	
cvar_0.5	10.0000	0.8732	0.8758	0.6137	
cvar_0.75	10.0000	0.8663	0.8783	0.6146	
cvar_0.9	10.0000	0.8663	0.8783	0.6146	
cvar_0.95	10.0000	0.8663	0.8783	0.6146	
centralized	10.0000	0.9078	0.8782	0.5851	
local_only	1.0000	0.9036	0.8690	0.4591	
grid_best_validated	10.0000	0.9013	0.8595	0.5981	
ga_best_validated	10.0000	0.8670	0.8839	0.6290	

Table 4: Final selected model clinical metrics

accuracy	balanced_accuracy	auroc	auprc	sensitivity	specificity	precision_ppv	npv	f1_death	
0.8403	0.8156	0.9055	0.6531	0.7837	0.8476	0.3981	0.9682	0.5280	0.

6 Training Dynamics

Figure 7: FedAvg loss across communication rounds.

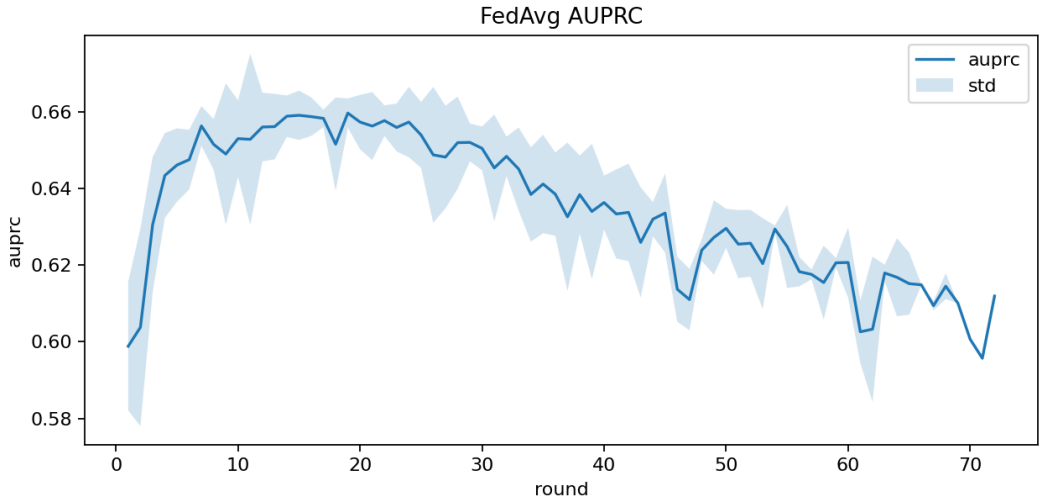

Figure 8: AUPRC across communication rounds.

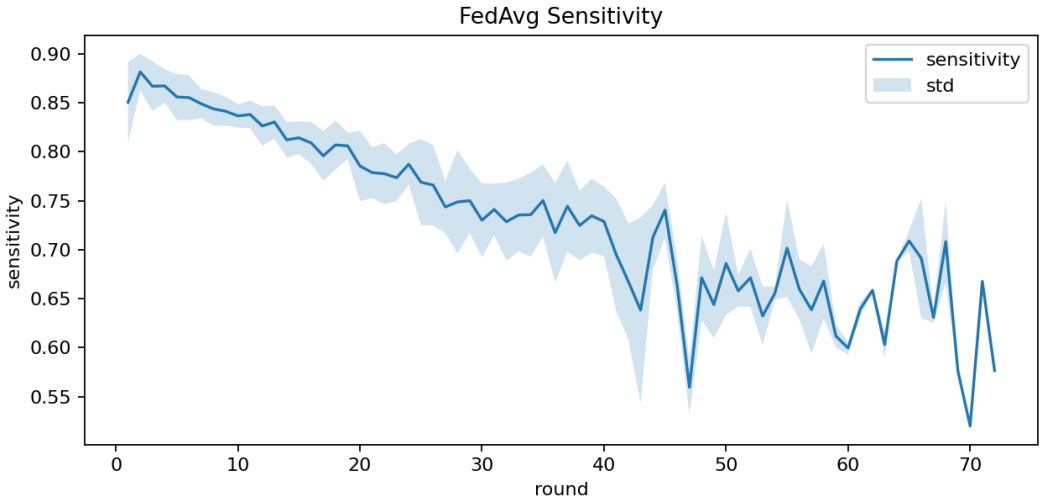

Figure 9: Mortality sensitivity across communication rounds.

Figure 10: Worst-client mortality recall across rounds.

7 Fairness and Client-Level Analysis

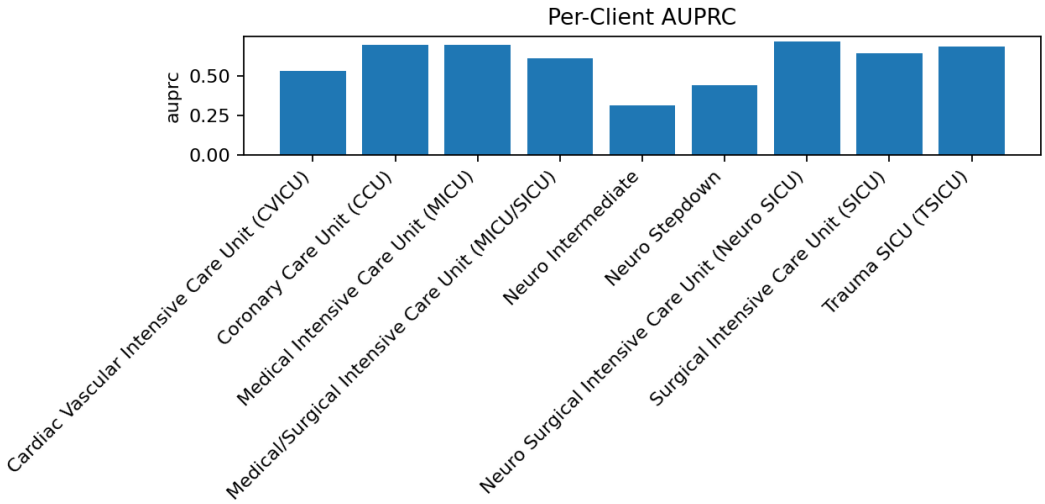

Figure 11: Per-client AUPRC for the final selected model.

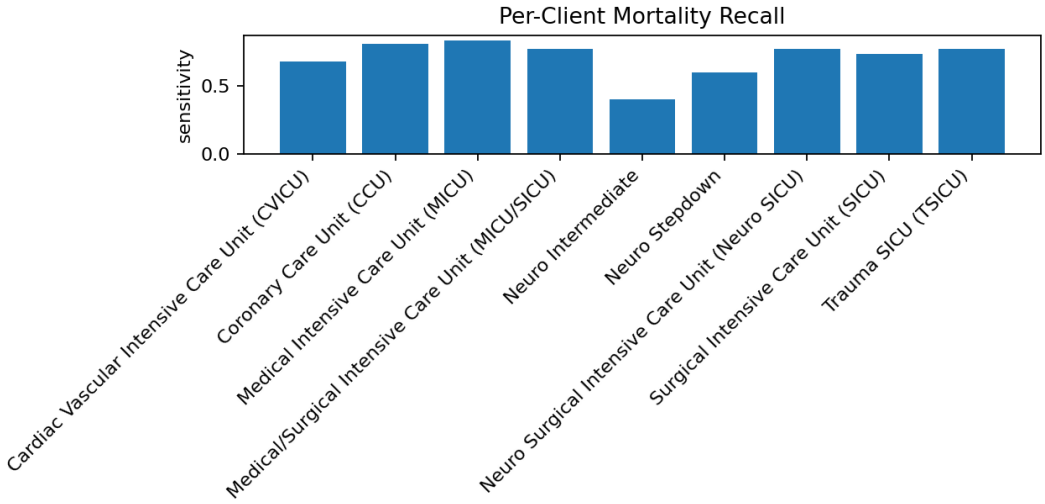

Figure 12: Per-client mortality recall for the final selected model.

Figure 13: Worst-client recall by method.

8 Clinical Classification

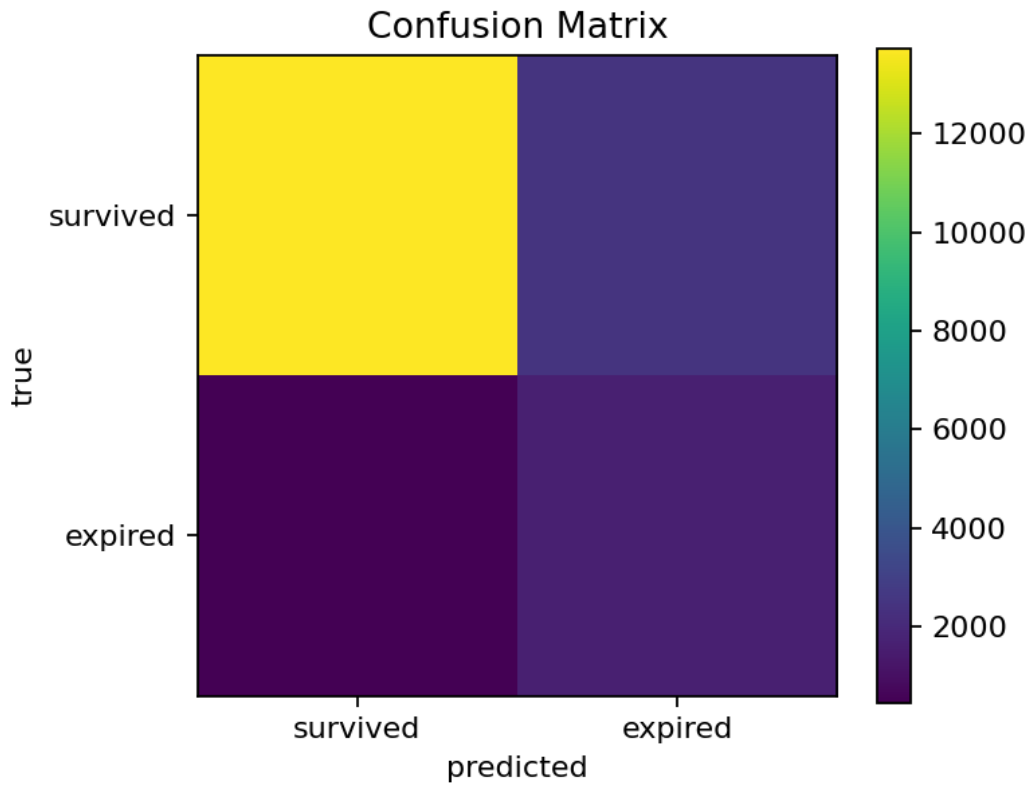

Figure 14: Confusion matrix for the final selected model.

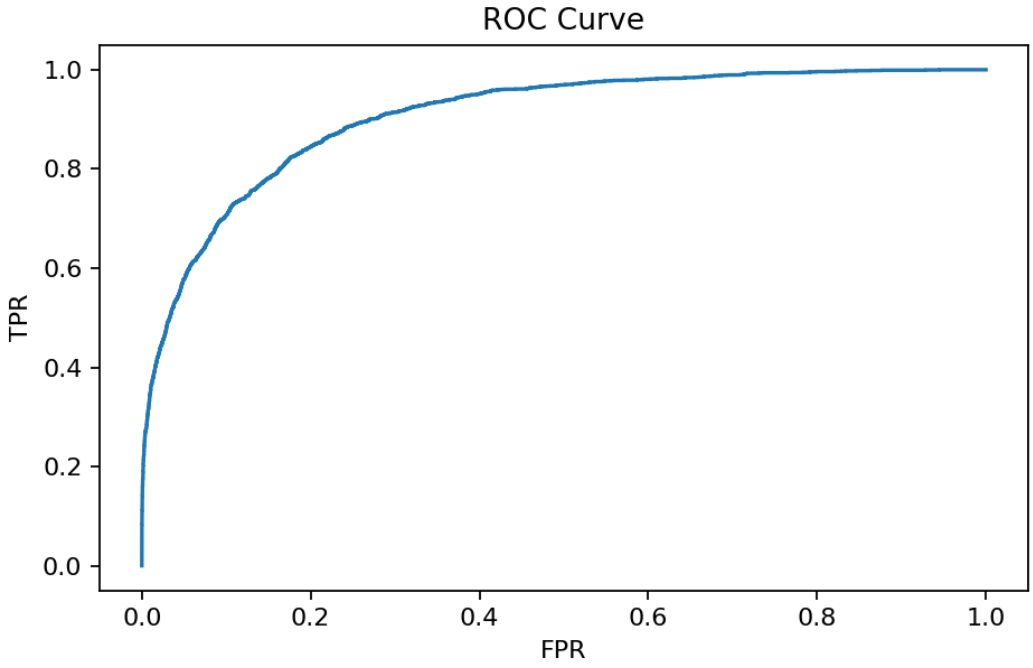

Figure 15: ROC curve.

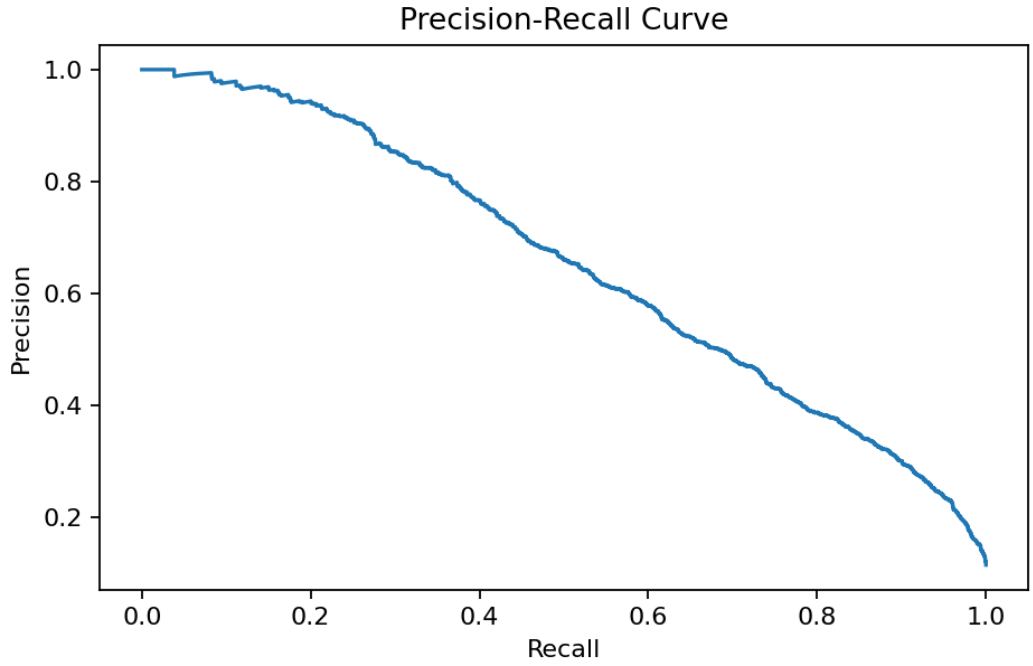

Figure 16: Precision-recall curve, which is especially important for imbalanced mortality prediction.

9 Search and Communication Optimization

The genetic search evaluated 48 candidate configurations. The best raw vector was `[1.2660918762649054, 3.4847698845669535, 0.005769185832593745]`. Grid and GA candidates were then validated with the same ten-seed protocol as FedAvg and CVaR.

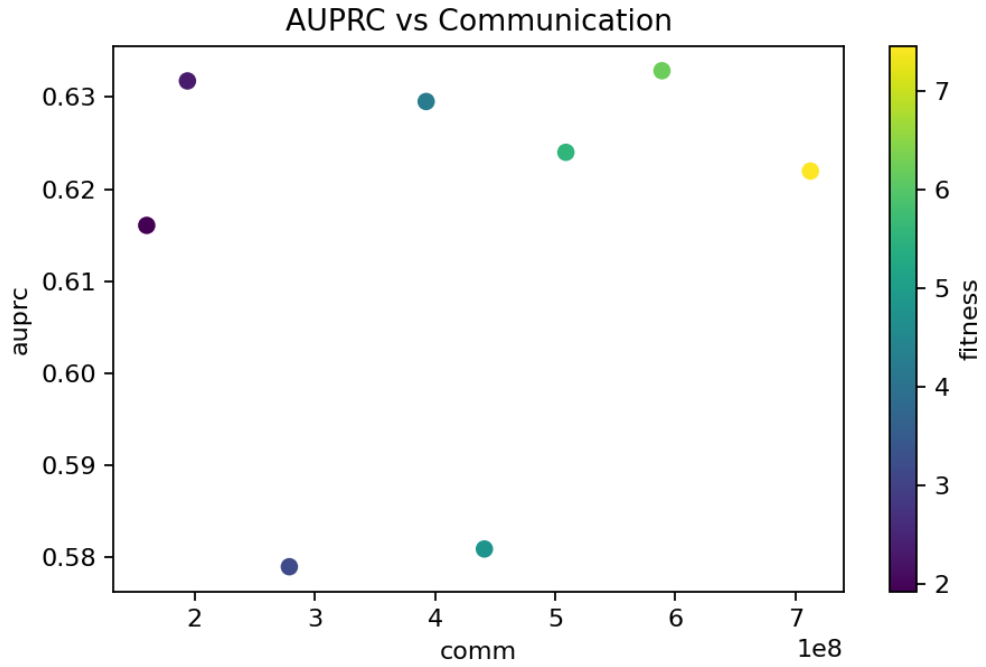

Figure 17: Search candidates plotted by AUPRC and communication.

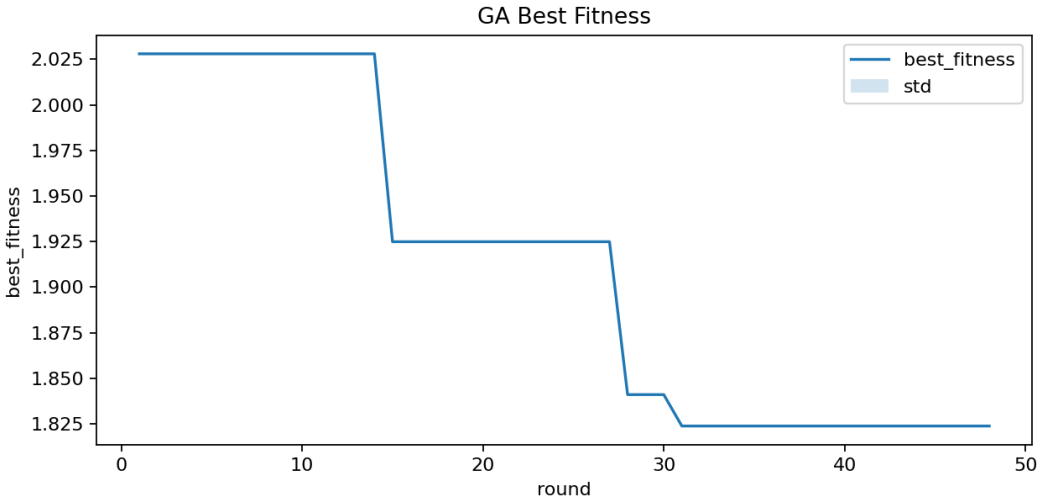

Figure 18: Best genetic-search fitness over evaluations.

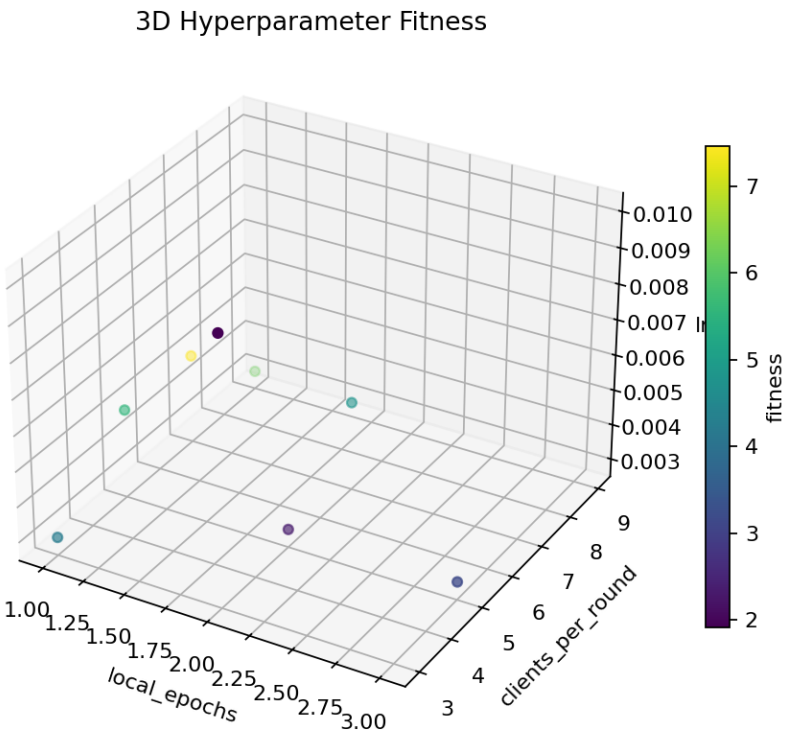

Figure 19: Three-dimensional hyperparameter search landscape.

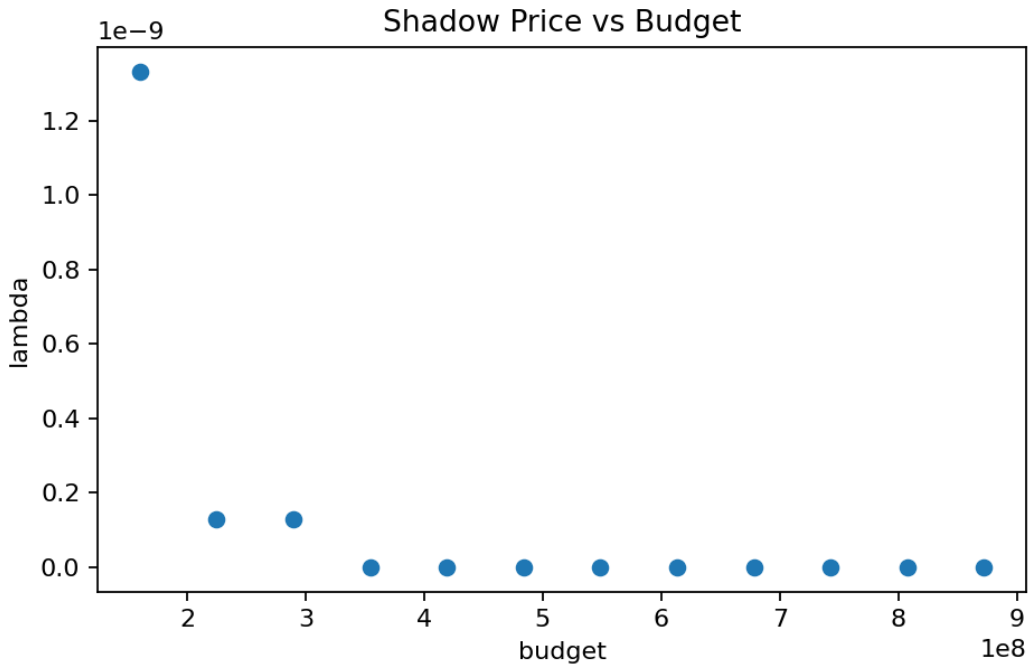

Figure 20: LP communication shadow price by budget.

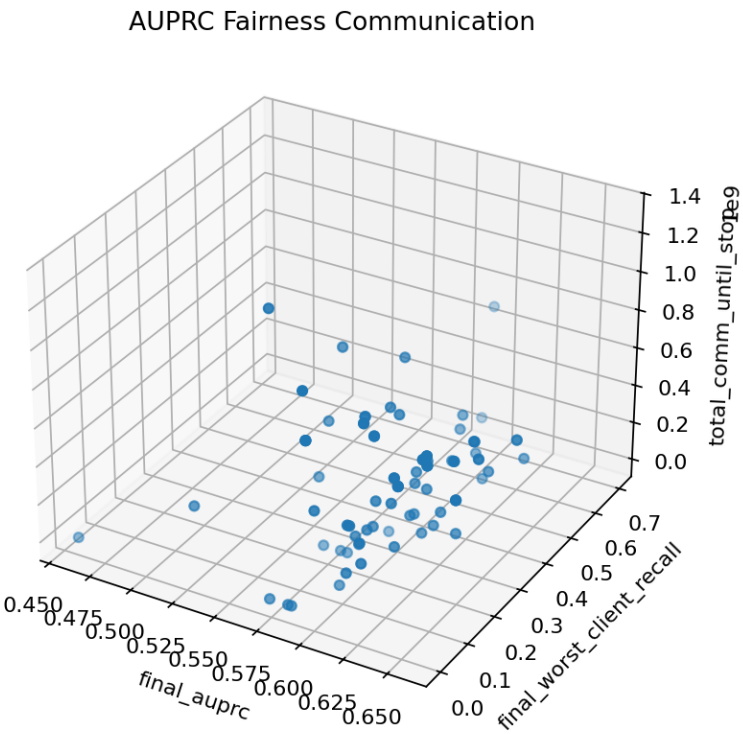

Figure 21: Three-way tradeoff between AUPRC, fairness, and communication.

10 Runtime

Table 5: Runtime by pipeline stage

stage	seconds
load_mimic_arrays	0.2641
fedavg_default	185.3920
cvar_sweep	989.0590
baselines	222.3467
search	1256.8964
predictions_metrics	9.6077
lp_diagnostics	0.1571
diagnostics	1.2190
plots	0.9817
artifacts	0.0200

11 Limitations

This run is a strong federated optimization baseline, but it should be interpreted carefully. Some administrative features may reflect coding processes rather than pure early physiology, so leakage-sensitive ablations should be used in future work. The local-only baseline is not directly comparable

to ten-seed global methods because it trains separate client models. Finally, high AUROC does not guarantee high death recall in every ICU unit, so false-negative analysis remains central.

12 Artifacts

The main output folder is `outputs/full_mimic_iv_training`. It contains raw round logs, method summaries, clinical metrics, predictions, calibration outputs, LP diagnostics, search history, plots, model checkpoints, and report drafts. The preprocessing folder `outputs/full_mimic_iv` contains the joined model matrix, EDA outputs, and the saved preprocessor.